Solving systems of equations with a variety of methods

1 The point of intersection of two lines.

2 -3

3 -4

4

a

x	$y = 3x - 1$	$y = -5x + 23$
1	2	18
2	5	13
3	8	8
4	11	3
5	14	-2

b $x = 3, y = 8$

5

a

x	$y = 5x$	$y = 9 - 4x$
1	5	5
3	15	-3
5	25	-11

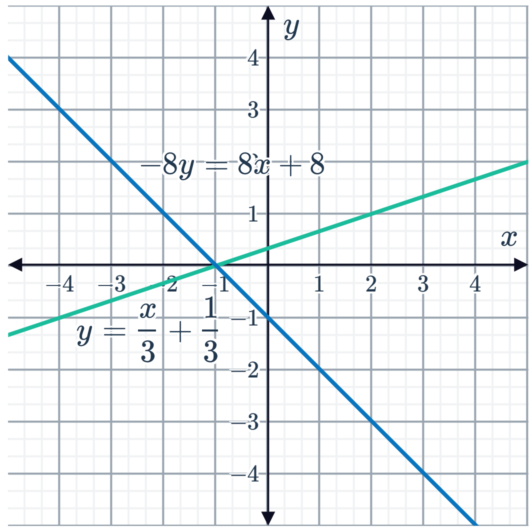
b $x = 1, y = 5$

6

a $(-1, 0), \left(0, \frac{1}{3}\right)$

b $(0, -1), (-1, 0)$

c



d $x = -1, y = 0$

7

a $x = 6, y = 4$

b $x = 0, y = -2$

c $x = 4, y = 0$

8

a $x = 8, y = 3$

b $x = \frac{20}{27}, y = \frac{62}{27}$

c $x = -20, y = 15$

d $x = 2, y = \frac{5}{3}$

e $p = \frac{2}{5}, q = \frac{3}{4}$

f $p = \frac{3}{5}, q = \frac{4}{5}$

9

a $x = 4, y = 2$

b Yes

c Yes

10 $x + y = 9, y - x = 1$

11

a $x = 13$

b $x = 9$

12

a 4

b $x = -3, y = -9$

c No

13

a $x = 1, y = 9$

b $y = -3x + 12$

c $3y = 3x + 24$

d $x = 1, y = 9$

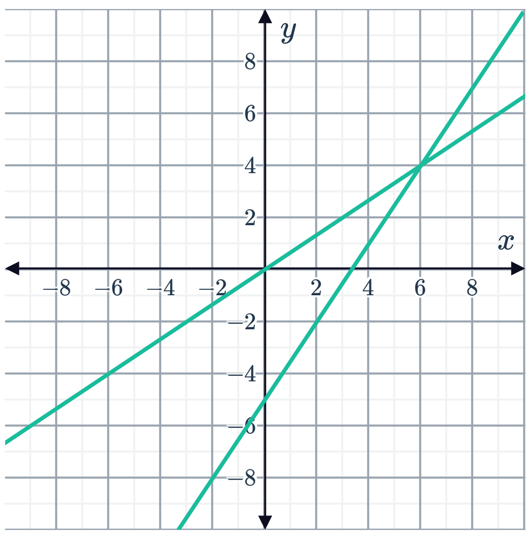
e Yes, Equations 3 and 4 are the result of applying identical operations to both the left hand side and the right hand side of each of the original equations.

14

a $x = 6$

b Equation 1 and Equation 4

c



d $x = 6$

15

a

i

x	1	3	5	7	9
y	58	64	70	76	82

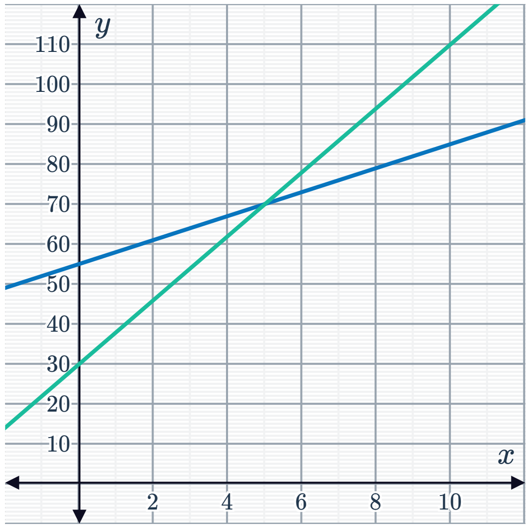
ii

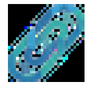
x	1	3	5	7	9
y	38	54	70	86	102

b

i No ii No iii Yes iv No

c





5 minutes

e TeleNow

16

a No **b** Yes **c** No **d** No

Additional questions

17

a $(0, 4)$ **b** $(4, 6)$ **c** $(8, 9)$ **d** $(-3, 4)$

e $(7, 2)$ **f** $(9, 6)$

18

a $(5, -1)$ **b** $(-2, 6)$ **c** No solutions **d** $(7, -6)$

19

a $(-3.125, 7.9375)$ **b** $\left(33, -\dfrac{100}{19}\right)$ **c** $\left(\dfrac{13}{3}, 36\right)$ **d** $\left(\dfrac{15147}{37}, \dfrac{347}{37}\right)$